

Zachary Benson

Senior full stack engineer (Java, Spring Boot, React, AWS, TypeScript, Angular, Mobile)

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PROFESSIONAL SUMMARY

As a software engineer with 8 years of experience, I bring strong technical expertise, a collaborative mindset, and a deep commitment to building high-quality software. I'm driven by curiosity and a constant desire to improve system performance, code quality, and overall engineering excellence.

I'm a highly motivated, results-oriented engineer who focuses on delivering meaningful outcomes. While I'm energized by achieving the final goal, I approach the process with discipline, adaptability, and a willingness to learn. I thrive in team environments that value open communication and shared ownership, yet I'm equally effective working independently with accountability and focus.

With a customer-centric perspective, I prioritize solving high-value problems and understanding the real needs behind each project. I think strategically about long-term technical direction, balancing engineering trade-offs with business objectives to create scalable, sustainable solutions.

As a problem-solver, I approach challenges analytically—identifying root causes, evaluating options, and developing practical, effective solutions. Across backend, frontend, and full-stack work, I bring a thoughtful, forward-looking approach that strengthens both the product and the team.

EDUCATION

University of Florida - Bachelor's Degree in computer Engineering, 2014 - June 2018

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TECHNICAL SKILLS

Backend & Microservices Architecture: Advanced expertise in Java, Kotlin, and Node.js (TypeScript) for building resilient, scalable systems. Proficient in Spring Boot, Spring Security, and Express.js, with deep experience in Microservices, BFF, Event-Driven Architecture, and Domain-Driven Design (DDD).

Mobile Engineering (Android Native): Specialist in Android development using Kotlin and Java. Expert in modern UI frameworks like Jetpack Compose and Material Design, utilizing MVVM/MVI and Clean Architecture to build robust, testable applications powered by Coroutines, Flow, and Hilt/Dagger.

API Design & Communication: Architected and implemented high-performance communication layers using REST, GraphQL, gRPC, and WebSockets. Skilled in managing asynchronous messaging and decoupling services via Kafka, RabbitMQ, and Google Cloud Pub/Sub.

Frontend & Interactive UIs: Proficient in building responsive, component-based web applications using React, Next.js, and Angular. Experienced in state management via Redux Toolkit and RxJS, ensuring top-tier UX with Tailwind CSS and Material UI.

Cloud Infrastructure & DevOps: Deep hands-on experience with AWS (Lambda, EC2, EKS, Fargate) and GCP (GKE, Apigee, BigQuery). Skilled in Infrastructure as Code (IaC) using Terraform and automating full-cycle delivery through Jenkins, GitHub Actions, and GitLab CI with ArgoCD for Kubernetes.

Database Systems & Persistence: Expert in designing and optimizing relational schemas (PostgreSQL, MySQL) and NoSQL solutions (MongoDB, DynamoDB). Proficient in high-performance caching with Redis and implementing advanced search capabilities via Elasticsearch and OpenSearch.

Security & Identity Management: Implemented enterprise-grade security protocols including OAuth 2.0, JWT, and OpenID Connect. Experienced in securing cloud environments using AWS Cognito, GCP Secret Manager, and RBAC patterns.

Testing, Quality & Observability: Committed to TDD/BDD practices using JUnit, Mockito, Espresso, and Cypress. Proactive in system health monitoring and debugging using Grafana, Datadog, Sentry, and Prometheus to ensure 99.9% uptime.

Agile Leadership & Collaboration: Strong background in Agile/Scrum methodologies, utilizing Jira and Confluence for streamlined project tracking. Experienced in leading code reviews and architectural discussions within globally distributed engineering teams.

WORK EXPERIENCE

Deloitte (Remote)- Java Full Stack Engineer

Jul 2024 – Present

- Architected and delivered a high-performance, **cloud-native e-commerce** platform using the Java stack (**Spring Boot, PostgreSQL, React**) within a **Microservices** architecture, ensuring scalability, modularity, and independent deployments via **Spring Cloud**.
- Led the design and architecture of **RESTful** and **GraphQL** APIs using Java and **Netflix DGS/Spring GraphQL**, establishing consistent patterns for versioning, error handling, and authentication (**Spring Security, JWT, OAuth2/OIDC**).
- Created a centralized API Gateway pattern with rate limiting and request validation using **Spring Cloud Gateway** and **Redis**, significantly improving latency and throughput across distributed services.
- Designed and built a highly scalable **React** component architecture using **TypeScript**, following atomic design principles and Storybook-driven development, supporting reusable, testable, and responsive UI components.
- Implemented **Next.js SSR/SSG** strategies, dynamic imports, and lazy loading to enhance frontend performance, **Core Web Vitals**, and **SEO** readiness for global **e-commerce traffic**.
- Defined and maintained backend modules with Spring Boot, applying **Domain-Driven Design (DDD)**, dependency injection, and **Spring Data JPA (Hibernate)** to improve maintainability and separation of concerns.
- Integrated secure payment workflows with **Stripe** and **PayPal**, ensuring PCI-DSS compliance, and built extensible Spring Interceptors and middleware for webhook handling and fraud detection.
- Built and deployed cloud infrastructure using AWS (App Runner, S3, RDS, Lambda) and Terraform for Infrastructure as Code (IaC), supporting high availability and disaster recovery.
- Automated full CI/CD pipelines using GitHub Actions and Jenkins, utilizing Docker, Helm, and Amazon EKS (Kubernetes) to enable canary releases and environment parity.
- Implemented full-stack observability using **Micrometer, Prometheus, Grafana**, and **ELK Stack**, establishing standardized dashboards and alerts for deep production insights.
- Applied **Test-Driven Development (TDD)** using JUnit 5, **Mockito, Cypress**, and **React Testing Library**, ensuring robust test coverage across unit, integration, and **E2E** flows.
- Collaborated with designers to translate Figma wireframes into responsive, accessible, and scalable UI systems, promoting consistent branding across platforms.
- Authored technical documentation, API specifications (**OpenAPI/Swagger**), and **DevOps** runbooks to support ongoing development and team knowledge sharing.
- Mentored frontend and backend engineers on Clean Architecture, code quality, and Agile/Scrum delivery practices, contributing to a high-performing, cross-functional team.

General Motors (Florida) – Java Full Stack Engineer

Oct 2022 - May 2024

- Architected and developed a real-time vehicle tracking platform using the **Java stack (Spring Boot, PostgreSQL, Angular)**, supporting thousands of concurrent users and delivering accurate live telemetry and fleet performance insights.
- Designed and deployed cloud-native infrastructure, leveraging **Amazon S3, AWS Lambda, API Gateway, and Amazon Cognito (or Spring Security OAuth2)**, ensuring secure authentication and highly available backend services.
- Engineered real-time data streaming systems using **Spring WebFlux** and **WebSockets**, enabling instant map updates, route tracking, and telemetry broadcasting with low-latency data handling in a distributed reactive environment.
- Developed secure authentication and authorization layers using **Spring Security, OAuth2, JWT, and Multi-Factor Authentication (MFA)**, enhancing system resilience and data protection.
- Built scalable, reusable UI components and Directives with **Angular + TypeScript**, implementing advanced state management patterns with **NgRx** and Services to ensure consistent and performant frontend behavior.
- Designed and implemented **RESTful APIs** and **GraphQL** endpoints (**Netflix DGS**) for modular data consumption, aligning backend services with scalable microservice communication models using Java 17/21.
- Optimized database performance using advanced indexing, partitioning, execution plan analysis, and Spring Data JPA (Hibernate), improving data retrieval speeds and reducing query load under high traffic.
- Implemented **CI/CD** pipelines using Jenkins, **GitLab CI**, and Docker, ensuring automated testing, build, and zero-downtime deployment via **Kubernetes (EKS)** across staging and production environments.
- Enabled infrastructure-as-code (IaC) practices using Terraform and **AWS CloudFormation**, standardizing environment provisioning and improving deployment consistency.
- Led development of interactive dashboards using custom charting libraries and Map APIs, utilizing **Angular's** change detection strategies and **RxJS** Observables for high-frequency visual analytics.
- Applied performance tuning techniques such as **Angular Route Guard lazy loading, OnPush** change detection, and distributed caching (**Redis**), significantly reducing initial load times and improving responsiveness.
- Authored and maintained comprehensive technical documentation, API references, architectural diagrams, and operational guides for development and **DevOps** teams.
- Championed **Agile** and **Scrum** methodologies, conducting sprint planning and code reviews while mentoring junior engineers in **SOLID** principles, **Clean Architecture**, and **Java best practices (Effective Java)**.
- Conducted extensive automated and manual testing using **JUnit 5, Mockito, Jasmine, and Karma**, ensuring robust **QA** coverage across backend and frontend services.

Zillow (Remote) – Java Full Stack Engineer

Mar 2020 - Sep 2022

- Designed and developed enterprise-level **real estate** platforms using the **Java stack (Spring Boot, PostgreSQL, Angular)**, delivering intuitive property search, listings, and management systems.
- **Architected** and deployed **cloud-native infrastructure**, leveraging **Amazon S3, AWS Lambda** (or **Google Cloud Functions**), and **API Gateway**, ensuring fault-tolerant, scalable **backend** services and efficient **serverless** data workflows.
- Developed advanced frontend features such as interactive map visualizations, dynamic filters, and real-time property listings, enhancing UX across web and mobile interfaces using **Angular** and **TypeScript**.
- Built and secured **RESTful APIs** for real-time data access and third-party integrations, including payment gateways, geolocation services, and MLS property data providers using Java 17/21.
- Implemented **SSR (Server-Side Rendering)** with **Angular** Universal and Spring Boot, incorporating lazy loading and intelligent caching strategies to significantly improve **SEO** and application performance.
- Optimized complex client-side state using **NgRx** and **RxJS**, enabling responsive UI behavior and consistent user session experiences across enterprise property dashboards.
- Integrated **TypeScript** and enforced modern development standards, improving maintainability, scalability, and type safety across the full **Spring Boot/Angular** stack.
- Collaborated with product owners and **UI/UX** teams under **Agile** and **Scrum** methodologies to prioritize backlogs, estimate workloads, and meet sprint deliverables within **Jira**.
- Led code reviews and provided technical mentorship on **Clean Architecture, SOLID** principles, and efficient component design patterns for both frontend and backend teams.
- Authored detailed system documentation, architectural flow diagrams, and deployment guides to support onboarding, scalability, and cross-team alignment.
- Automated testing workflows using **JUnit 5, Mockito, Jasmine**, and **Karma**, ensuring high code coverage, regression protection, and reliable production releases.
- Initiated performance audits and implemented database indexing, query optimization via Spring Data JPA (**Hibernate**), and **CDN** integration, reducing latency across key workflows.

TePe USA (California) – Full Stack Engineer

Sep 2018 – Feb 2020

- Architected high-scale enterprise healthcare platforms using **Angular and TypeScript**, focusing on reusable UI components, **modular design patterns**, and **Angular Feature Modules**.
- Leveraged **Angular Material** for consistent CSS styling; engineered complex Menus and Drawer components using TypeScript for strict type safety and **reactive component communication**.
- Integrated the **html2canvas** package within the Angular ecosystem to provide advanced client-side control over image rendering and screen capture functionality.
- Implemented robust identity management and secure user authentication flows utilizing **Spring Security** and **Auth0**.
- Engineered high-performance backend services using **Java and Spring Boot (Web API)**, adhering to **Domain-Driven Design (DDD)** and **SOLID** principles for clean separation of concerns.
- Managed relational data schemas with **PostgreSQL** (or MySQL) and utilized **Spring Data JPA (Hibernate)** for database control and object-relational mapping (ORM).
- Streamlined data transfer using **MapStruct** (or ModelMapper) for DTO mapping; integrated **SpringDoc OpenAPI** to generate interactive **Swagger UI** documentation.
- Conducted rigorous code quality audits and performance profiling using **SonarQube** and browser profiling tools to optimize frontend delivery and backend efficiency.
- Managed the full deployment lifecycle for frontend and backend projects hosted on **DigitalOcean**, utilizing droplets and **Docker** for high availability.
- Applied **TDD principles** using **JUnit 5, Mockito, and Jasmine/Karma** to ensure high code coverage for critical business logic across the full stack